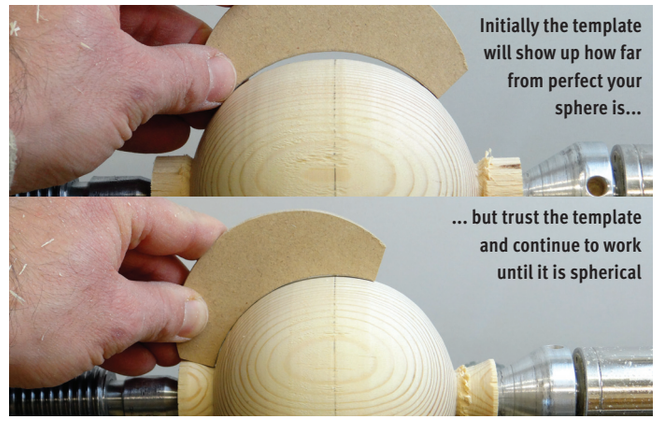
Start checking

As the globe shape emerges from your blank you can bring the template into play. As long as you paid close attention to making the template and did it accurately, you can trust the template. If you are not going to believe what it's telling you, there was no point in making it! By now your orb should be looking quite spherical, but the chances are your template will tell you it isn't. Take the blank out of the lathe and turn it on its end so the poles are top and bottom. Iit is often surprising how this will show you how spherical your ball isn't... yet! Little by little, cut by cut, remove small amounts of timber as your template guides you and soon enough, you should have a nearly perfect sphere with a relatively small block of waste at each end. With the shaping process, the waste blocks will naturally reduce in size, to the point where they are greatly reduced but still securely driving the blank.



Initially the template will show up how far from perfect your sphere is...

... but trust the template and continue to work until it is spherical

Removing the waste

Removing the waste with a hand saw

There are several ways that you could remove the waste, but the safest way is to saw it off with a hand saw. Parting on the lathe would only remove the waste from one end so you would still need to saw off the other waste block anyway and using a bandsaw could potentially be dangerous – if the blade grabbed the wood and spun the sphere it could endanger your fingers.

With the waste removed all that is left to do is to tidy up the cut marks, but potentially the sphere might not be 100% perfect, so it would be useful to be able to remount the ball on the lathe in such a way that it would allow the removal of the cut areas and to correct any imperfections in the first turning and shaping stage... well there is a way, and this is the beauty of this method of making spheres and the part that makes it foolproof. In addition, it is an easy method to follow as well as inexpensive to create the parts to make it work.

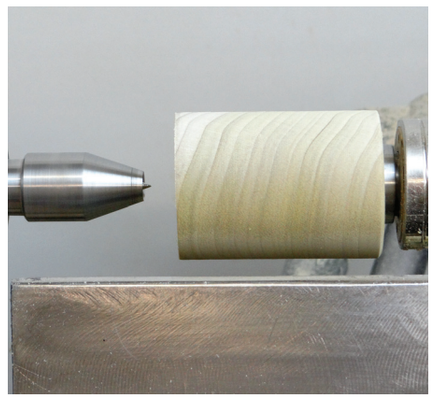
Wooden mounting cups

To achieve this secondary holding method we need to turn some wooden cups to hold the sphere securely without marking it. I chose two pieces of tulipwood (*Liriodendron tulipifera*), a relatively soft hardwood of around the same density as the redwood I used for the balls. One is simply held in the chuck and turned to a cup shape, the other needs to fit onto your live centre. I am fortunate that I own a Oneway live centre which features a threaded portion, designed to hold various attachments and homemade holding devices such as the wooden cup needed here. These centres are acknowledged to be one of the best on the market, but have a price tag to match. There are now some very similar versions available.

To make accessories for this style of live centre, you need to buy a tap that matches, in this case a ¾in x 10tpi UNC, but check your centre to find the correct size. Any wood can be tapped; evidence of this is the fact that I have used this soft tulipwood here, but you need to select a piece of side grain – like a bowl blank, rather than like a spindle – as this will hold the thread, whereas the threads cut into an end grain blank will just crumble, unless you use a very dense wood, but this wouldn't be suitable in this case.

I cut a square of timber, oversized in all directions, pilot drilled a hole and tapped out the thread. I'm sure an engineer would wince at my tapping technique, but it does the job! With the thread cut, I fit the block onto the live centre. With a drive centre in the headstock, I bring the tailstock up and engage the centres until the block spins and I turn it to a cylinder. This can then be removed from the live centre, reversed, and held in the chuck to turn the cup shape into the face.

The tapped hole in the cross grain blank of tulipwood



The blank is mounted on the live centre and simply driven with a drive centre

Truing the blank for the cup with a spindle gouge

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